

PRAGUNA MANVI | RESUME

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SUMMARY

Software engineer passionate about solving real-world challenges; currently building deep learning quantization tools at MathWorks.

EXPERIENCE

SOFTWARE ENGINEER - MATHWORKS

5/2024 -
Bangalore

I currently work as a Software Engineer in Deep Learning, developing optimized quantization algorithms for the MATLAB's deep learning framework to enable inference using fixed-point compute on resource-constrained targets. I also work on building utilities for code generation workflows.

SENIOR SDE - MAI LABS

6/2023 - 5/2024
Bangalore/Hyderabad

Led a real-time analytics team focused on enterprise blockchain transactions, building a scalable, distributed platform to flag suspicious activity and visualize KYT (Know Your Transaction) data. Also, developed an LLM-based platform with a custom prompt architecture for vulnerability analysis of customer-uploaded smart contracts.

INTERN - INTEL HABANA AI

8/2022 - 3/2023
Bangalore

During this internship, contributed to Habana Pytorch's software stack for enabling eager mode Deep learning training and inference on Habana's accelerators. The role included rewriting custom torch-ops to the graph compiler's frontend, where a few core tensor operators were implemented and added to the custom torch plugin.

SOFTWARE ENGINEER - DAILYHUNT

2/2020 - 10/2021
Bangalore

Developed highly scalable systems (>10k rps), which includes an efficient distributed caching and storage system for managing metadata and hashtags, a concurrent notification system for millions of users, a high-load backend for social group management, and a customized feed generation system for user activities on the Dailyhunt platform.

EDUCATION

IIIT HYDERABAD - MS BY RESEARCH

9/2021 - 11/2023

Graduate Research Assistant at the Center for Visual Information Technology (CVIT) under Prof. Anoop Namboodiri. The research work has been predominantly in vision, deep learning, cryptography, biometrics, and template protection systems with a CGPA of **9.0**.

B.M.S COLLEGE OF ENGINEERING - B.E

8/2016 - 8/2020

Graduated in Computer Science and Engineering with **9.45** CGPA.

SKILLS

PROGRAMMING LANGUAGE
BACKEND FRAMEWORKS
FRONTEND FRAMEWORKS
DEEPLARNING LIBRARIES
LANGUAGES

Python | Java | C++ | Kotlin | GoLang | JavaScript | MATLAB
spring-boot | gin | express | fast-api
React | Angular
Pytorch | Tensorflow
English | Kannada | Hindi

PUBLICATIONS / PATENTS / ACHIEVEMENTS

- Presented "SIAN: Secure Iris Authentication Using Noise" at IJCB (International Joint Conference on Biometrics) 2022 - proposed a novel cryptography theory with provable information-theoretic security and a computationally efficient computer vision application for secure verification.
- Provisional Patent Filed titled, "SYSTEM AND METHOD FOR AUTHENTICATING AN IRIS OF A USER USING BIOMETRIC NOISE". U.S patent pending.
- The second work titled, "S-BAN: Secure Biometric Authentication using Noise," accepted in ICVGIP 2023, an extension of the previous paper with additional novel algorithmic utilities and a deep learning architecture design for all biometric modalities.
- Finalist of India Police Hackathon 2019: Face Recognition App to detect missing individuals on the private police database.
- Runner up in Intel's HyperLedger Hackathon 2019: Securing Medical Records using Blockchain